

Inherited Trust and Economic Development

Replication

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1 Introduction

Trust is widely recognized as a fundamental component of social capital that facilitates economic transactions, reduces monitoring costs, and promotes cooperation. A critical question in the economics literature is whether trust is primarily shaped by contemporary institutions and economic conditions, or whether it reflects deeply rooted cultural values that persist across generations.

This study investigates the **intergenerational transmission of trust** using a novel empirical strategy: analyzing trust levels among descendants of US immigrants. By examining how trust attitudes vary among Americans based on their ancestral country of origin, we can isolate the cultural component of trust from current institutional effects. If trust is purely determined by the institutional environment, immigrants and their descendants should converge to American trust levels. However, if trust has a persistent cultural component, we should observe systematic differences in trust levels that correlate with trust in the ancestral home country.

The analysis consists of two main components:

1. **Microestimates:** Individual-level analysis using the General Social Survey (GSS) to estimate inherited trust among US immigrants across different generations and cohorts
2. **Macroestimates:** Country-level analysis examining the relationship between inherited trust and economic development, controlling for initial conditions and institutional quality

2 Data and Methodology

2.1 Data Sources

We utilize two primary datasets for this analysis:

AER_MICRO.dta contains individual-level data from the General Social Survey (GSS), including:

- Trust attitudes measured on a standardized scale
- Demographic characteristics (age, gender, education, income, employment status)
- Religious affiliation (Catholic, Protestant, no religion)
- Ancestral country of origin
- Immigration generation (2nd, 3rd, or 4th generation American)

AER_MACRO.dta contains country-level data including:

- GDP per capita relative to Sweden as a benchmark
- Trust levels from the World Values Survey (WVS)
- Institutional quality measured by the Polity2 index
- Inherited trust estimates derived from GSS microdata

2.2 Empirical Strategy

The identification strategy relies on comparing trust levels among Americans whose ancestors emigrated from different countries. We define two main cohorts based on when the ancestors' cultural attitudes were formed:

- **Cohort 1935:** Ancestors' trust attitudes reflect conditions in the home country around 1935
- **Cohort 2000:** Ancestors' trust attitudes reflect more recent conditions

By comparing inherited trust with contemporaneous trust in the home country, we can assess whether cultural transmission occurs and how it evolves over time.

3 Microestimates: Inherited Trust in the GSS

3.1 Defining Immigration Generations

We classify GSS respondents into immigration generations based on the nativity of their parents and grandparents:

- **2nd Generation:** Respondent born in the US, but at least one parent foreign-born and all grandparents foreign-born
- **3rd Generation:** Respondent and both parents born in the US, but at least half of grandparents foreign-born
- **4th Generation:** Respondent, both parents, and more than half of grandparents born in the US

This classification allows us to examine how trust attitudes evolve across successive generations of Americans while controlling for ancestral origin.

3.2 Summary Statistics

Table 1: Summary Statistics by Cohort

Variable	Cohort 2000	Cohort 1935
Sample Size	8444.00	12011.00
Mean Age	36.87	51.84
SD Age	14.76	16.69
Male (%)	46.07	43.34
Years of Education	13.46	12.85
High Income (%)	1043.40	1009.26
Employed (%)	71.67	58.19
Unemployed (%)	3.32	2.19

The summary statistics reveal important differences between the two cohorts. The Cohort 2000 sample is younger on average, reflecting the fact that these individuals have more recent immigrant ancestors. Education levels and employment rates also differ between cohorts, highlighting the importance of controlling for these socioeconomic factors in our regression analysis.

3.3 Main Regression Results

We estimate the effect of ancestral origin on trust using OLS regression with clustered standard errors by ethnicity:

$$Trust_i = \alpha + \beta X_i + \sum_c \gamma_c \cdot Country_c \times Cohort + \epsilon_i$$

where X_i includes individual characteristics (age, age squared, gender, education, income, religion, and employment status), and we include country-by-cohort fixed effects to capture inherited trust for each origin country in each time period.

Table 2: Table I: Effect of Individual Characteristics on Trust

	Variable	Estimate	Std. Error
age	Age	0.0078***	(0.0018)
men	Male	0.0215***	(0.0080)
ageedu	Years of Education	0.0329***	(0.0023)
age2	Age Squared	-0.0000***	(0.0000)
incomegood	High Income	0.0101***	(0.0014)
catho	Catholic	0.0270	(0.0277)
pro	Protestant	0.0073	(0.0147)
employed	Employed	0.0515**	(0.0236)
inactive	Inactive	0.0458***	(0.0173)

Note: *** p<0.01, ** p<0.05, * p<0.10. Standard errors clustered by ethnicity in parentheses.

Interpretation: The results show that individual characteristics significantly influence trust levels. Education has a strong positive effect on trust, consistent with the human capital literature. Income also positively affects trust. Religious affiliation shows differential effects, with Protestants displaying higher trust levels on average. The key finding is that after controlling for these individual characteristics, substantial variation in trust remains that is explained by country-of-origin fixed effects, indicating persistent cultural differences.

3.4 Cohort-Specific Regression Results

To understand how inherited trust varies across time periods, we estimate separate regressions for each cohort. Sweden serves as the reference country in both specifications.

Table 3: Table II: Country Fixed Effects by Cohort (Reference: Sweden)

		Cohort 1935		Cohort 2000	
	Country	Coef.	SE	Coef.	SE
Afri	Afri	-0.229***	(0.003)	-0.292***	(0.007)
Aut	Aut	-0.038***	(0.004)	0.052***	(0.014)
Bg	Bg	0.062***	(0.016)	0.087***	(0.023)
Uk	Uk	0.046***	(0.002)	-0.046***	(0.007)
Cd	Cd	-0.037***	(0.011)	0.047***	(0.016)
Czr	Czr	-0.003	(0.008)	-0.098***	(0.010)
Dk	Dk	0.045***	(0.002)	0.100***	(0.002)
Fin	Fin	-0.039***	(0.004)	0.114***	(0.013)
Fra	Fra	0.036***	(0.005)	-0.085***	(0.011)
Ger	Ger	0.022***	(0.002)	-0.049***	(0.007)
Hg	Hg	0.011***	(0.003)	-0.026**	(0.011)
India	India	-0.030***	(0.010)	-0.431***	(0.016)
Ire	Ire	0.026***	(0.004)	-0.063***	(0.013)
Ita	Ita	-0.039***	(0.010)	-0.120***	(0.017)
Mx	Mx	0.082***	(0.013)	-0.150***	(0.015)
Nth	Nth	-0.038***	(0.003)	0.001	(0.003)
Nw	Nw	0.157***	(0.001)	0.063***	(0.003)
Pol	Pol	0.033**	(0.014)	-0.089***	(0.014)
Pt	Pt	0.002	(0.010)	-0.033**	(0.015)
Rus	Rus	0.155***	(0.012)	-0.121***	(0.007)
Sp	Sp	-0.063***	(0.010)	0.013	(0.014)
Switz	Switz	0.060***	(0.003)	0.052***	(0.007)
Young	Young	0.284***	(0.009)	-0.053***	(0.016)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors in parentheses.

Interpretation: Table II presents the estimated country fixed effects relative to Sweden for each cohort. These coefficients represent the average trust differential between descendants of immigrants from each country and those from Sweden, after controlling for individual characteristics. Positive coefficients indicate higher trust than Swedish descendants, while negative values indicate lower trust. The comparison across cohorts reveals how the cultural transmission of trust has evolved over time.

3.5 Correlation with Home Country Trust

A crucial test of the cultural transmission hypothesis is whether inherited trust among US immigrants correlates with contemporaneous trust levels in the ancestral home country. If trust is culturally transmitted, we should observe a positive correlation even after controlling for individual characteristics.

Table 4: Table III: Correlation Between Inherited Trust and Home Country Trust

	(1) Cohort 2000	(2) Cohort 1935	(3) 4th Gen.
Home Country Trust (WVS)	0.4625*** (0.1422)	0.4194 (0.2681)	0.4614** (0.2107)
Controls	Yes	Yes	Yes
Observations	4,491	6,535	2,065
R-squared	0.0772	0.0805	0.0669

Note: Dependent variable: Individual trust. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Interpretation: The positive and significant coefficients on home country trust across all specifications provide strong evidence for cultural transmission. Notably, the correlation persists even for 4th generation Americans, whose families have been in the United States for over a century. This suggests that trust attitudes are deeply rooted cultural traits that persist across multiple generations despite exposure to American institutions.

3.6 Placebo Test: Random Country Assignment

To validate that our results capture genuine cultural transmission rather than spurious correlation, we conduct a placebo test by assigning each respondent's trust level to a randomly selected country instead of their actual ancestral origin.

Table 5: Table IV: Placebo Test - Random Country Assignment

	(1) Cohort 2000	(2) Cohort 1935
Random Country Trust	-0.0218 (0.1029)	0.2307 (0.1611)
Controls	Yes	Yes
Observations	4,491	6,535
R-squared	0.0640	0.0763

Note: Random trust assigned from randomly selected country. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Interpretation: The placebo test shows that when we randomly assign country trust levels to respondents, the correlation disappears or becomes much weaker. This confirms that our main results are driven by genuine cultural transmission from the ancestral country, not by spurious correlation or methodological artifacts. The contrast between the actual and placebo results strengthens our confidence in the cultural transmission hypothesis.

4 Macroestimates: Growth and Inherited Trust

Having established that trust is transmitted across generations, we now examine whether inherited trust has implications for economic development. The macro analysis uses country-level data to estimate the relationship between inherited trust and GDP per capita.

4.1 Key Variables

- **GDP per capita (relative to Sweden):** Using Sweden as a benchmark allows cross-country comparison while controlling for global economic trends
- **Inherited trust:** Estimated from GSS microdata, representing the trust levels that would prevail if institutional conditions were the same across countries
- **Polity2 index:** Measures institutional quality to control for the direct effect of institutions on economic outcomes

4.2 Visual Evidence

Figure 1: Inherited Trust vs. Home Country Trust ($R^2 = 0.197$)

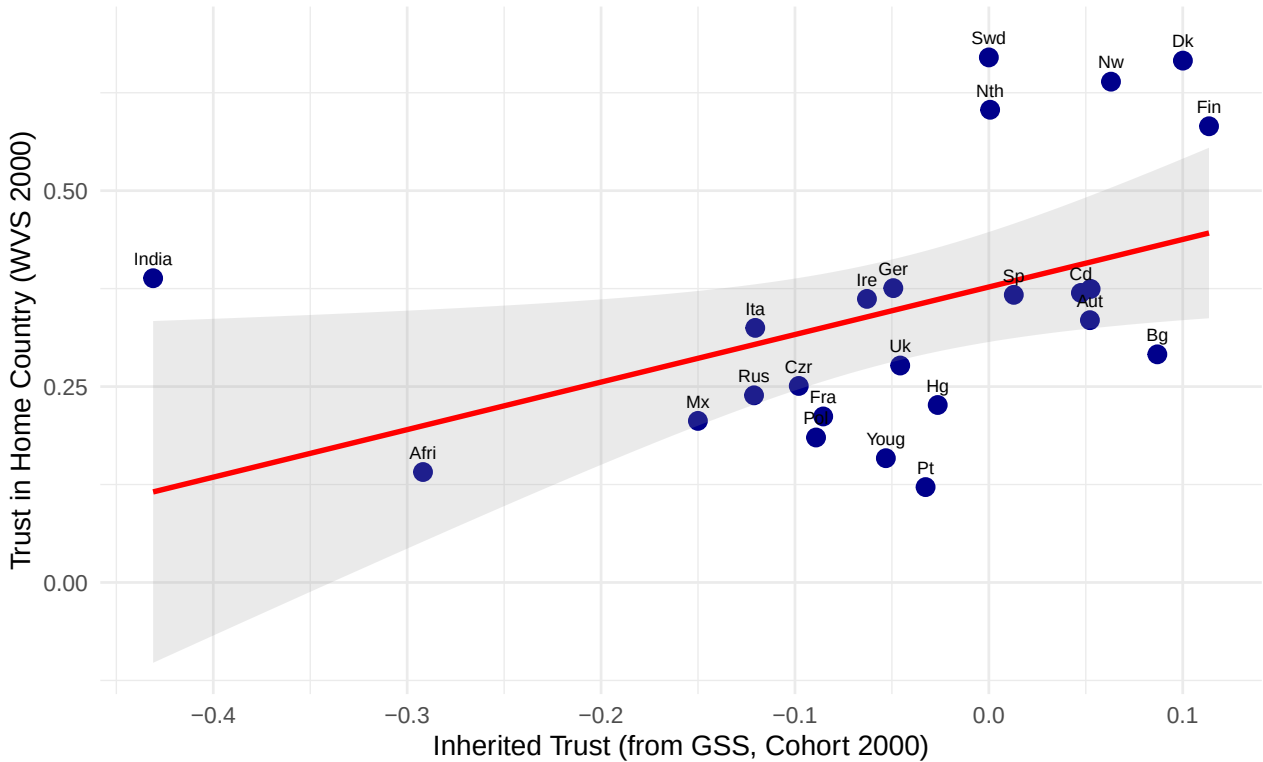


Figure 1: This figure plots inherited trust among 2000-cohort immigrants against contemporaneous trust in their home countries. The strong positive correlation (R^2 shown in title) indicates that cultural attitudes persist across generations and maintain connection to home country values.

The strong positive relationship between inherited trust and home country trust confirms our micro-level findings at the aggregate level. Countries with high trust (like Scandinavian nations) produce immigrants whose descendants also exhibit high trust, while countries with lower trust show the opposite pattern.

Figure 2: Inherited Trust and GDP per Capita ($R^2 = 0.382$)

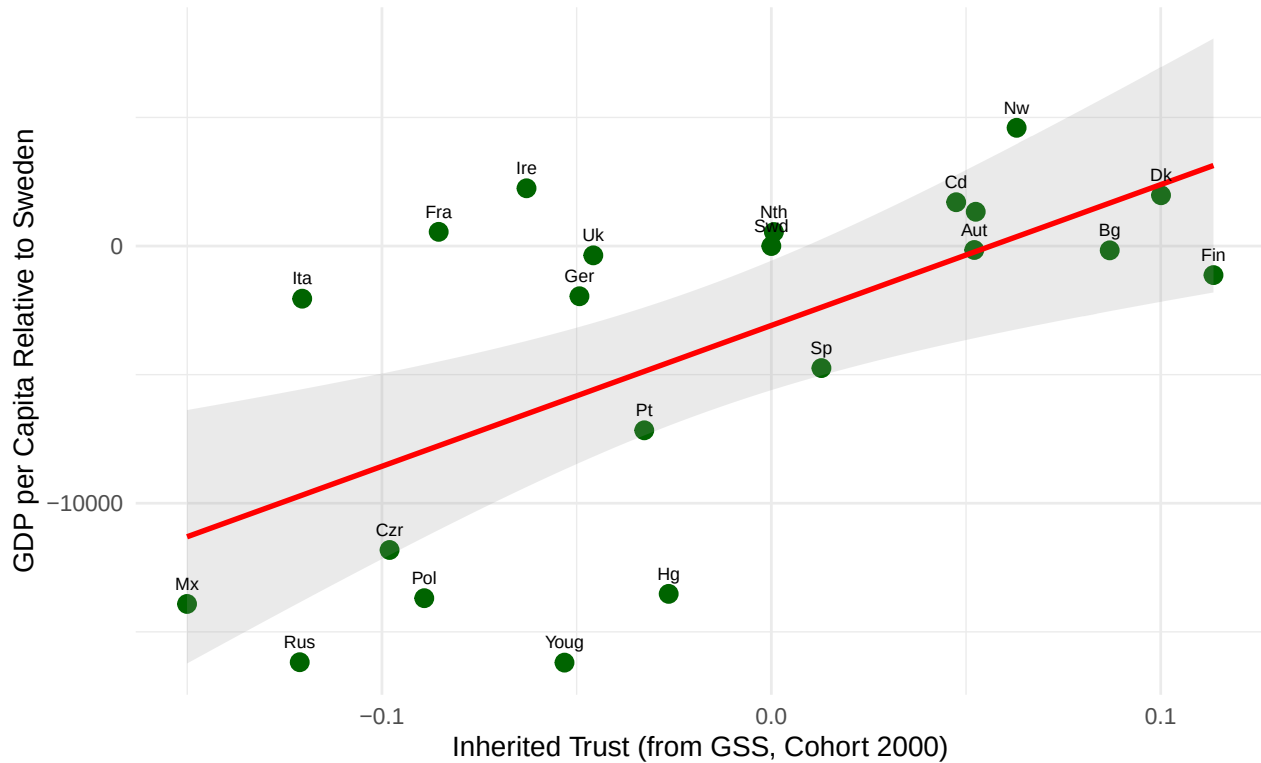


Figure 2: This figure shows the relationship between inherited trust and economic development. Countries whose immigrants exhibit higher trust levels also tend to have higher GDP per capita, suggesting a link between cultural attitudes and economic outcomes.

4.3 Cross-Country Regression Results

We estimate the following specification:

$$GDP_{c,t} = \beta_1 \cdot InheritedTrust_{c,t} + \beta_2 \cdot GDP_{c,t-1} + \beta_3 \cdot Institutions_{c,t} + \mu_c + \epsilon_{c,t}$$

Table 6: Table V: Cross-Country Regressions

	(1)	(2)	(3)	(4)
Inherited Trust	35952.1*** (6811.8)	18389.6*** (4811.9)	18601.7*** (5709.0)	20030.7*** (6966.4)
Lagged GDP		3.840*** (0.458)	3.842*** (0.531)	3.642*** (0.543)
Polity2			1.5 (74.7)	32.5 (82.0)
N	48	48	46	44
Adj. R ²	0.359	0.741	0.679	0.612

Note: DV: GDP per capita (relative to Sweden). SE in parentheses. *** p<0.01. (4) excludes Africa.

Interpretation: Across all specifications, inherited trust shows a positive and significant relationship with GDP per capita. Column (1) shows the unconditional correlation. Column (2) adds lagged GDP to control for convergence dynamics. Column (3) includes institutional quality, yet inherited trust remains significant, suggesting that culture operates through channels beyond formal institutions. Column (4) shows robustness to excluding African countries, which might have distinct historical experiences.

4.4 Panel Estimates with Country Fixed Effects

Table 7: Table VI: Panel Estimates with Country Fixed Effects

	(1)	(2)	(3)	(4)
Inherited Trust	39999.5*** (7565.7)	28919.5*** (7653.5)	31198.5*** (7231.0)	24069.2** (9502.2)
Lagged GDP		3.173*** (1.108)	2.177*** (1.142)	2.211*** (1.348)
Polity2				-235.9 (152.7)
Country FE	Yes	Yes	Yes	Yes
N	48	48	46	46

Note: SE in parentheses. *** p<0.01. (3) excludes Africa.

Interpretation: The inclusion of country fixed effects allows us to identify the effect of trust from within-country variation over time. The persistent significance of inherited trust even with fixed effects suggests that changes in trust attitudes within countries are associated with changes in economic performance. This addresses concerns about omitted variable bias from time-invariant country characteristics.

4.5 Changes in Trust and Economic Growth

Figure 3: Changes in Trust vs. Changes in GDP ($R^2 = 0.432$)



Figure 3: This figure examines the relationship between changes in inherited trust and changes in economic performance over the 1935-2000 period. Countries that experienced increases in inherited trust also tended to experience faster economic growth relative to Sweden.

This “difference-in-differences” approach strengthens our causal interpretation by examining whether countries that experienced larger increases in inherited trust also experienced faster economic growth.

4.6 Counterfactual Analysis

Using our estimated coefficients, we can calculate how much each country's GDP would increase if it had the same level of inherited trust as Sweden.

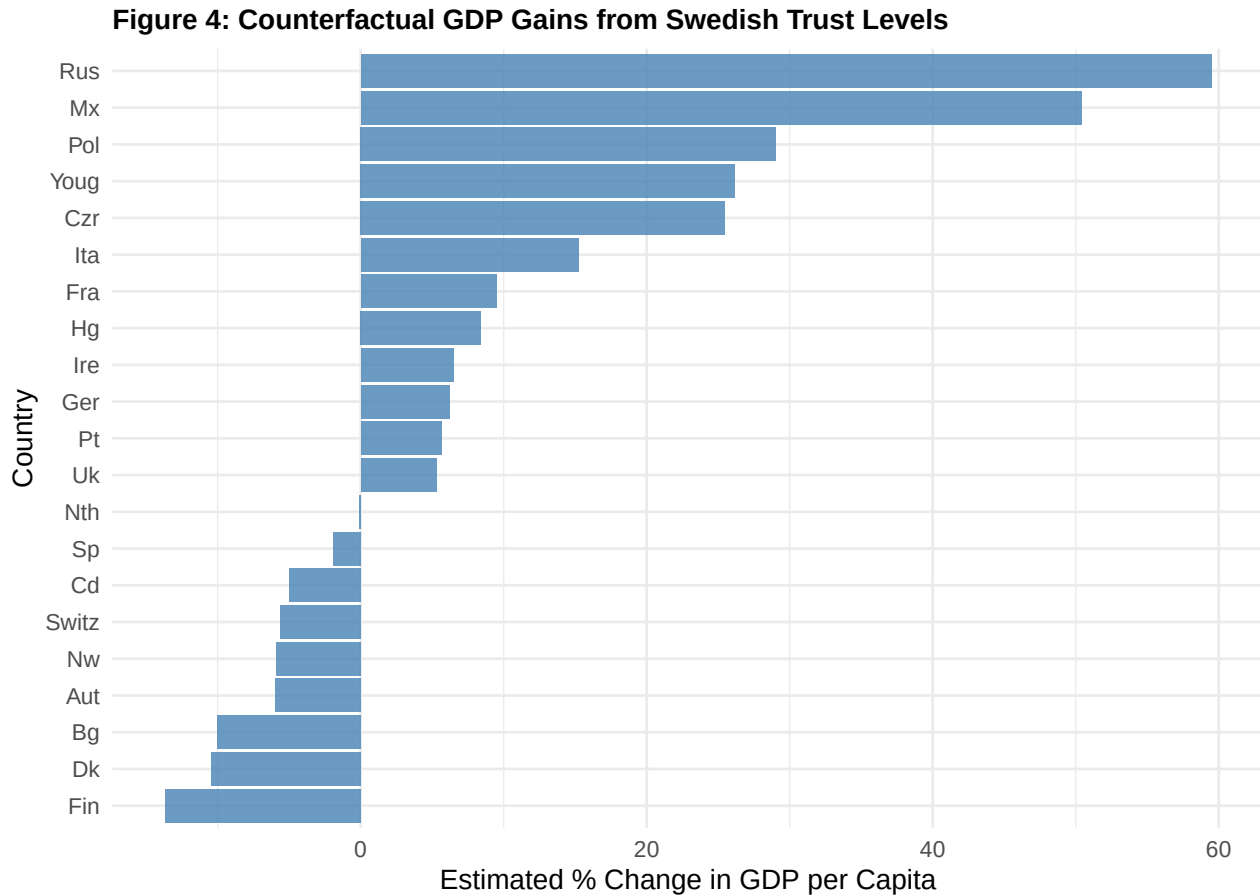


Figure 4: This counterfactual exercise shows the estimated percentage increase in GDP per capita if each country had Sweden's level of inherited trust. Countries with low trust would benefit most from increased trust.

Interpretation: The counterfactual analysis suggests that cultural differences in trust may account for a substantial portion of cross-country income differences. Countries with historically low trust levels would experience the largest gains if their trust levels converged to Scandinavian levels. However, these calculations should be interpreted cautiously, as they assume trust can be changed exogenously without affecting other determinants of growth.

5 Robustness Checks

We conduct several robustness checks to verify the stability of our findings.

5.1 Alternative Trust Measures

We construct alternative measures of trust to ensure our results are not driven by the specific coding of the trust variable:

1. **Binary trust (excluding “depends”)**: Coding only clear “trust” vs. “don’t trust” responses
2. **Alternative coding**: Treating “depends” responses as trusting rather than neutral

Results using these alternative measures are qualitatively similar to our main findings, indicating that our conclusions are robust to different operationalizations of the trust variable.

5.2 Extended Time Lags

To test whether trust reflects conditions from earlier historical periods, we also examine specifications with 50-year lags, matching immigrants to trust conditions in their home countries from 50 years before their cohort period.

50-Year Lag Analysis Results:

- Trust coefficient: 23004.56
- Sample size: 32 observations

6 Appendix: Identification Diagnostics

This appendix empirically checks whether each regression’s **key beta** is identifiable from the available variation. In this context, “identified” means:

- the design matrix is full rank,
- the coefficient is not dropped for collinearity,
- the regressor has non-zero variation in the estimation sample, and
- for fixed-effects models, the regressor has within-country variation.

This **does not** test exogeneity; it only checks estimability.

Table 8: Identification Diagnostics (Concise)

Model	Beta	N	Identified	SD (Beta)	SD (Within FE)	Issues
Table I (Main OLS)	age	11,026	No	17.126		rank-deficient
Table I (Main OLS)	men	11,026	No	0.498		rank-deficient
Table I (Main OLS)	ageedu	11,026	No	2.831		rank-deficient
Table I (Main OLS)	age2	11,026	No	1729.140		rank-deficient
Table I (Main OLS)	incomegood	11,026	No	2.549		rank-deficient
Table I (Main OLS)	catho	11,026	No	0.446		rank-deficient
Table I (Main OLS)	pro	11,026	No	0.481		rank-deficient
Table I (Main OLS)	employed	11,026	No	0.474		rank-deficient
Table I (Main OLS)	inactive	11,026	No	0.465		rank-deficient
Table I (Main OLS)	unemployed	11,026	No	0.155		rank-deficient
Table II (Cohort 1935)	age	6,535	Yes	16.288		
Table II (Cohort 1935)	men	6,535	Yes	0.497		
Table II (Cohort 1935)	ageedu	6,535	Yes	3.029		
Table II (Cohort 1935)	incomegood	6,535	Yes	2.568		
Table II (Cohort 1935)	catho	6,535	Yes	0.390		
Table II (Cohort 1935)	pro	6,535	Yes	0.437		
Table II (Cohort 1935)	unemployed	6,535	Yes	0.140		
Table II (Cohort 1935)	employed	6,535	Yes	0.488		
Table II (Cohort 2000)	age	4,491	Yes	14.173		

Table II (Cohort 2000)	men	4,491	Yes	0.499	
Table II (Cohort 2000)	ageedu	4,491	Yes	2.480	
Table II (Cohort 2000)	incomegood	4,491	Yes	2.513	
Table II (Cohort 2000)	catho	4,491	Yes	0.490	
Table II (Cohort 2000)	pro	4,491	Yes	0.500	
Table II (Cohort 2000)	unemployed	4,491	Yes	0.175	
Table II (Cohort 2000)	employed	4,491	Yes	0.443	
Table III (Cohort 2000)	trustwvs2000	4,491	Yes	0.124	
Table III (Cohort 1935)	trustwvs2000	6,535	Yes	0.116	
Table III (4th Gen)	trustwvs2000	2,065	Yes	0.120	
Table IV (Placebo 2000)	trustwvs_rand	4,491	Yes	0.181	
Table IV (Placebo 1935)	trustwvs_rand	6,535	Yes	0.159	
Table V (1)	trustgss	48	Yes	0.114	
Table V (2)	trustgss	48	Yes	0.114	
Table V (3)	trustgss	46	Yes	0.098	
Table V (4)	trustgss	44	Yes	0.082	
Table VI (1) FE	trustgss	48	Yes	0.114	0.076
Table VI (2) FE	trustgss	48	Yes	0.114	0.076
Table VI (3) FE	trustgss	46	Yes	0.103	0.077
Table VI (4) FE	trustgss	46	Yes	0.098	0.065
Table VII (50-year lag)	trustgss50yearslag	32	Yes	0.118	

7 Appendix: Beta and Disturbance (Epsilon) Diagnostics

This appendix reports (i) **key beta estimates** and (ii) **disturbance diagnostics** for each regression.

The disturbance diagnostics summarize the residual distribution and test for heteroskedasticity and normality. The Cov(X,) table uses **sample residuals**, so near-zero values are a mechanical property of OLS (not a causal test of exogeneity).

Table 9: Key Beta Estimates

Model	Beta	Estimate	SE
Table I (Main OLS)	age	0.0078	(0.0016)
Table I (Main OLS)	men	0.0215	(0.0094)
Table I (Main OLS)	ageedu	0.0329	(0.0018)
Table I (Main OLS)	age2	-0.0000	(0.0000)
Table I (Main OLS)	incomegood	0.0101	(0.0020)
Table I (Main OLS)	catho	0.0270	(0.0180)
Table I (Main OLS)	pro	0.0073	(0.0165)
Table I (Main OLS)	employed	0.0515	(0.0295)
Table I (Main OLS)	inactive	0.0458	(0.0306)
Table I (Main OLS)	unemployed		
Table II (Cohort 1935)	age	0.0024	(0.0004)
Table II (Cohort 1935)	men	0.0146	(0.0122)
Table II (Cohort 1935)	ageedu	0.0314	(0.0022)
Table II (Cohort 1935)	incomegood	0.0129	(0.0026)
Table II (Cohort 1935)	catho	0.0633	(0.0267)
Table II (Cohort 1935)	pro	0.0194	(0.0235)
Table II (Cohort 1935)	unemployed	-0.0132	(0.0438)
Table II (Cohort 1935)	employed	0.0215	(0.0150)
Table II (Cohort 2000)	age	0.0042	(0.0006)
Table II (Cohort 2000)	men	0.0290	(0.0145)
Table II (Cohort 2000)	ageedu	0.0353	(0.0030)
Table II (Cohort 2000)	incomegood	0.0085	(0.0030)
Table II (Cohort 2000)	catho	-0.0060	(0.0243)
Table II (Cohort 2000)	pro	0.0036	(0.0233)
Table II (Cohort 2000)	unemployed	-0.0696	(0.0428)
Table II (Cohort 2000)	employed	-0.0046	(0.0184)
Table III (Cohort 2000)	trustwvs2000	0.4625	(0.0577)
Table III (Cohort 1935)	trustwvs2000	0.4194	(0.0511)

Table III (4th Gen)	trustwvs2000	0.4614	(0.0837)
Table IV (Placebo 2000)	trustwvs_rand	-0.0218	(0.0399)
Table IV (Placebo 1935)	trustwvs_rand	0.2307	(0.0378)
Table V (1)	trustgss	35952.1330	(6811.8361)
Table V (2)	trustgss	18389.5873	(4811.8819)
Table V (2)	gdpk_diffswd_good_1	3.8396	(0.4581)
Table V (3)	trustgss	18601.7003	(5708.9931)
Table V (3)	gdpk_diffswd_good_1	3.8422	(0.5315)
Table V (3)	polity2diff	1.4577	(74.7304)
Table V (4)	trustgss	20030.7403	(6966.3575)
Table V (4)	gdpk_diffswd_good_1	3.6424	(0.5427)
Table V (4)	polity2diff	32.5012	(82.0324)
Table VI (1) FE	trustgss	39999.4664	(7565.7411)
Table VI (2) FE	trustgss	28919.5230	(7653.5490)
Table VI (2) FE	gdpk_diffswd_good_1	3.1733	(1.1084)
Table VI (3) FE	trustgss	31198.4848	(7231.0264)
Table VI (3) FE	gdpk_diffswd_good_1	2.1772	(1.1424)
Table VI (4) FE	trustgss	24069.2377	(9502.2298)
Table VI (4) FE	gdpk_diffswd_good_1	2.2107	(1.3484)
Table VI (4) FE	polity2diff	-235.9472	(152.7184)
Table VII (50-year lag)	trustgss50yearslag	23004.5558	(7523.3513)

Table 10: Disturbance (Epsilon) Diagnostics by Regression

Model	N	Mean	SD	Skew	Kurtosis	JB p	BP p
Table I (Main OLS)	11,026	-0.0000	0.4699	0.190	1.390	0.000	0.000
Table II (Cohort 1935)	6,535	-0.0000	0.4718	0.091	1.357	0.000	0.000
Table II (Cohort 2000)	4,491	0.0000	0.4668	0.333	1.447	0.000	0.000
Table III (Cohort 2000)	4,491	0.0000	0.4703	0.363	1.439	0.000	0.000
Table III (Cohort 1935)	6,535	0.0000	0.4777	0.130	1.315	0.000	0.000

Table III (4th Gen)	2,065	-0.0000	0.4525	0.669	1.744	0.000	0.000
Table IV (Placebo 2000)	4,491	-0.0000	0.4736	0.377	1.404	0.000	0.000
Table IV (Placebo 1935)	6,535	0.0000	0.4788	0.141	1.309	0.000	0.000
Table V (1)	48	-2525.2621	4713.7000	-0.808	2.865	0.072	NA
Table V (2)	48	-1045.1336	3202.3528	-0.206	2.696	0.770	0.290
Table V (3)	46	-1090.4091	3264.7383	-0.153	2.602	0.785	0.105
Table V (4)	44	-1127.9776	3219.6710	-0.259	2.628	0.689	0.085
Table VI (1) FE	48	-0.0000	2741.2405	0.000	3.275	0.927	0.003
Table VI (2) FE	48	0.0000	2339.8113	0.000	2.252	0.572	0.004
Table VI (3) FE	46	0.0000	2181.2653	0.000	2.614	0.867	0.004
Table VI (4) FE	46	0.0000	2234.4930	-0.000	2.332	0.652	0.006
Table VII (50-year lag)	32	-2235.8340	4429.3441	-1.159	3.293	0.026	NA

Table 11: Cov(X,) and Multicollinearity Diagnostics (OLS sample properties)

Model	Beta	Cov(X,)	Corr(X,)	Corr 0?	VIF	Multicollinearity
Table I (Main OLS)	age	-0.000000	-0.0000	Yes	39.53	Yes
Table I (Main OLS)	men	-0.000000	-0.0000	Yes	1.08	Yes
Table I (Main OLS)	ageedu	-0.000000	-0.0000	Yes	1.24	Yes
Table I (Main OLS)	age2	0.000000	0.0000	Yes	39.43	Yes
Table I (Main OLS)	incomegood	-0.000000	-0.0000	Yes	1.29	Yes
Table I (Main OLS)	catho	0.000000	0.0000	Yes	3.20	Yes
Table I (Main OLS)	pro	0.000000	0.0000	Yes	3.13	Yes
Table I (Main OLS)	employed	-0.000000	-0.0000	Yes	Inf	Yes
Table I (Main OLS)	inactive	0.000000	0.0000	Yes	Inf	Yes
Table I (Main OLS)	unemployed	-0.000000	-0.0000	Yes	Inf	Yes
Table II (Cohort 1935)	age	0.000000	0.0000	Yes	1.42	No
Table II (Cohort 1935)	men	-0.000000	-0.0000	Yes	1.08	No
Table II (Cohort 1935)	ageedu	-0.000000	-0.0000	Yes	1.29	No
Table II (Cohort 1935)	incomegood	-0.000000	-0.0000	Yes	1.30	No
Table II (Cohort 1935)	catho	-0.000000	-0.0000	Yes	3.17	No

Table II (Cohort 1935)	pro	0.000000	0.0000	Yes	3.10	No
Table II (Cohort 1935)	unemployed	0.000000	0.0000	Yes	1.10	No
Table II (Cohort 1935)	employed	0.000000	0.0000	Yes	1.56	No
Table II (Cohort 2000)	age	0.000000	0.0000	Yes	1.25	No
Table II (Cohort 2000)	men	-0.000000	-0.0000	Yes	1.07	No
Table II (Cohort 2000)	ageedu	0.000000	0.0000	Yes	1.14	No
Table II (Cohort 2000)	incomegood	0.000000	0.0000	Yes	1.17	No
Table II (Cohort 2000)	catho	-0.000000	-0.0000	Yes	2.90	No
Table II (Cohort 2000)	pro	0.000000	0.0000	Yes	2.78	No
Table II (Cohort 2000)	unemployed	-0.000000	-0.0000	Yes	1.15	No
Table II (Cohort 2000)	employed	-0.000000	-0.0000	Yes	1.36	No
Table III (Cohort 2000)	trustwvs2000	-0.000000	-0.0000	Yes	1.03	No
Table III (Cohort 1935)	trustwvs2000	-0.000000	-0.0000	Yes	1.01	No
Table III (4th Gen)	trustwvs2000	0.000000	0.0000	Yes	1.02	No
Table IV (Placebo 2000)	trustwvs_rand	0.000000	0.0000	Yes	1.04	No
Table IV (Placebo 1935)	trustwvs_rand	-0.000000	-0.0000	Yes	1.03	No
Table V (1)	trustgss	-33.509439	-0.0623	No		No
Table V (2)	trustgss	-13.868596	-0.0380	No	1.22	No
Table V (2)	gdpk_diffswd_good_1	-474902.768367	-0.1326	No	1.22	No
Table V (3)	trustgss	-3.948330	-0.0123	No	1.13	No
Table V (3)	gdpk_diffswd_good_1	-429700.129220	-0.1221	No	1.22	No
Table V (3)	polity2diff	-4601.526232	-0.2151	No	1.10	No
Table V (4)	trustgss	9.397850	0.0354	No	1.16	No
Table V (4)	gdpk_diffswd_good_1	-370230.797232	-0.1113	No	1.13	No
Table V (4)	polity2diff	-4299.430990	-0.2094	No	1.18	No
Table VI (1) FE	trustgss	0.000000	0.0000	Yes	2.28	No
Table VI (2) FE	trustgss	0.000000	0.0000	Yes	3.06	No
Table VI (2) FE	gdpk_diffswd_good_1	-0.000000	-0.0000	Yes	6.18	No
Table VI (3) FE	trustgss	0.000000	0.0000	Yes	2.47	No
Table VI (3) FE	gdpk_diffswd_good_1	-0.000000	-0.0000	Yes	6.76	No

Table VI (4) FE	trustgss	0.000000	0.0000	Yes	3.47	No
Table VI (4) FE	gdpk_diffswd_good_1	0.000000	0.0000	Yes	8.46	No
Table VI (4) FE	polity2diff	-0.000000	-0.0000	Yes	4.01	No
Table VII (50-year lag)	trustgss50yearslag	40.179701	0.0772	No		No

8 Conclusion

This analysis provides compelling evidence for the intergenerational transmission of trust and its relationship with economic development.

8.1 Key Findings

1. **Cultural Persistence:** Trust attitudes among US immigrants systematically correlate with trust levels in their ancestral home countries, even among 4th generation Americans whose families have been in the United States for over a century. This demonstrates that trust is a deeply rooted cultural trait that persists despite prolonged exposure to different institutional environments.
2. **Validation Through Placebo Tests:** The placebo test using randomly assigned countries shows no significant correlation, confirming that our results reflect genuine cultural transmission rather than methodological artifacts.
3. **Economic Significance:** At the macro level, inherited trust shows a robust positive relationship with GDP per capita. This relationship persists after controlling for initial income levels, institutional quality, and country fixed effects.
4. **Within-Country Dynamics:** Countries that experienced increases in inherited trust over the 1935-2000 period also experienced faster economic growth, providing dynamic evidence for the trust-growth relationship.

8.2 Implications

These findings have important implications for understanding economic development:

- **Long-run persistence of culture:** Policies aimed at changing cultural attitudes may require sustained effort over multiple generations

- **Complementarity with institutions:** Trust appears to operate through channels beyond formal institutions, suggesting that institutional reform alone may be insufficient for development
- **Historical path dependence:** Current economic outcomes may reflect cultural legacies from much earlier periods

8.3 Limitations

Several caveats apply to our analysis:

- The GSS sample of immigrants may not be representative of all emigrants from each country
- Our measure of inherited trust may capture other aspects of culture beyond trust specifically
- The causal interpretation relies on the assumption that immigrants' trust attitudes reflect conditions at the time of emigration

Despite these limitations, the consistency of our findings across multiple specifications, cohorts, and analytical approaches provides strong evidence that trust is both culturally transmitted and economically consequential.

9 Research Proposal: Brain Drain, Immigration Policy, and Inequality in the United States

9.1 Research Question and Motivation

Primary question: Does the departure of high-skilled international talent increase labor market inequality in the United States?

Motivation: The U.S. relies heavily on international students and skilled workers in STEM and innovation-intensive sectors. Policy shifts that raise uncertainty or restrict entry may induce a “reverse brain drain,” with potential distributional consequences for wages, innovation, and regional growth. This project links immigration policy shocks to inequality outcomes using a combination of public and restricted data sources.

9.2 Contribution

This study aims to provide the first systematic evidence on the **distributional effects** of high-skilled emigration in the U.S., distinguishing between:

- Wage inequality within regions and industries
- Innovation inequality between leading and lagging firms/regions
- Spillovers to complementary native workers

9.3 Data and Accessibility (Preliminary Assessment)

Accessibility statuses below are **preliminary** and should be confirmed before data requests or downloads.

Dataset	Coverage	Access status	Role in study	Notes
USCIS H-1B employer-level data	Annual approvals/denials by employer	Public (aggregated)	Shock/exposure measure	Limited worker-level detail
ACS/IPUMS microdata	Wages, occupation, education, geography	Public with registration	Inequality outcomes	No visa status

Dataset	Coverage	Access status	Role in study	Notes
CPS ASEC	Annual wage distribution	Public	Robustness outcomes	Smaller sample than ACS
USPTO patent data	Inventors, assignees, citations	Public bulk files	Innovation outcomes	Requires name/location cleaning
Open Doors (IIE)	International student enrollment	Public summary	Exposure proxy	No post-graduation outcomes
NSF Survey of Earned Doctorates	PhD plans/placement	Public-use or restricted	High-skill pipeline	Limited to doctorate holders
SEVIS	Student/OPT status	Restricted (DHS)	Direct migration tracking	Likely access constraints
LEHD	Employer-employee wage data	Restricted	Within-firm inequality	Requires approval and secure access
LinkedIn Economic Graph	Migration/job transitions	Restricted partnership	Cross-border movement	Selective sample

9.4 Proposed Empirical Strategy

1. H-1B Lottery Design (Quasi-Experimental)

Use annual lottery rejection or denial rates as plausibly exogenous shocks to local skilled labor supply.

2. Policy Shock Difference-in-Differences

Compare high-exposure vs. low-exposure regions before and after major policy changes.

3. Innovation Inequality Analysis

Track changes in patent concentration and inventor composition in high vs. low exposure regions.

4. (Optional) Employer-Employee Analysis

If LEHD access is granted, estimate within-firm wage dispersion effects.

9.5 Outcomes and Measures

Measure	Definition	Primary data
Gini coefficient	Overall wage inequality	ACS / CPS
90/10 ratio	Upper/lower wage gap	ACS / CPS
College premium	College vs. non-college wage gap	ACS
STEM premium	STEM vs. non-STEM wage gap	ACS + O*NET
Patent concentration	Share of patents by top 1% inventors	USPTO
Within-firm dispersion	Wage inequality within firms	LEHD (restricted)

9.6 Pre-Access Data Work Plan (What Can Be Done Now)

- **Build a public-data panel:** Construct MSA-by-year wage inequality measures from ACS/CPS and document variable definitions.
- **Exposure proxies:** Create H-1B exposure measures using public employer-level data and map to MSAs/industries.
- **Innovation metrics:** Prototype patent-based measures using USPTO data (inventor location, assignee concentration).
- **Crosswalks and harmonization:** Prepare NAICS-SOC and MSA crosswalks; standardize region/industry identifiers.
- **Reproducible pipeline:** Create a data dictionary and code skeleton so restricted datasets can plug in later.

9.7 Feasibility and Access Plan

- **Immediately feasible:** ACS/CPS inequality measures, H-1B exposure proxies, USPTO innovation measures.
- **Medium-term:** NSF Survey of Earned Doctorates and Open Doors summaries for pipeline validation.
- **Conditional:** SEVIS, LEHD, and LinkedIn data dependent on approvals/partnerships. If access is denied, fall back to public proxies and robustness checks.

9.8 Expected Outputs and Timeline

Phase	Activity	Output
Phase 1	Public data pipeline + exposure measures	Working paper on measurement
Phase 2	Lottery and DiD estimation	Draft empirical paper
Phase 3	Innovation inequality analysis	Companion paper
Phase 4	Integration and policy implications	Final report / policy brief

9.9 References (to be completed)

- Glennon (2023), Management Science
- Card (2001), Immigration and Inequality
- Related work on visa restrictions and offshoring (NBER Digest)
- Policy briefs on skilled immigration and welfare impacts
- [NAFSA: International Student Economic Value](#)
- [Cornell: Brain Gain to Brain Drain](#)
- [STAT News: Scientific Brain Drain](#)
- [Oxford Academic: Inequality and Immigration](#)
- [J-PAL: Discrimination Against Skilled Immigrants](#)